

CSE327

Software Engineering

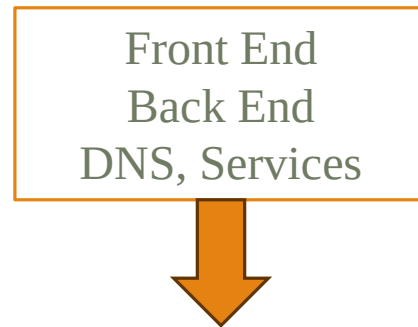
AKM IQTIDAR NEWAZ [IQN]

LECTURE - 7

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Agile

- Let's think about waterfall model for a big application like banking app, insurance application. Here, a huge application consists of small parts.

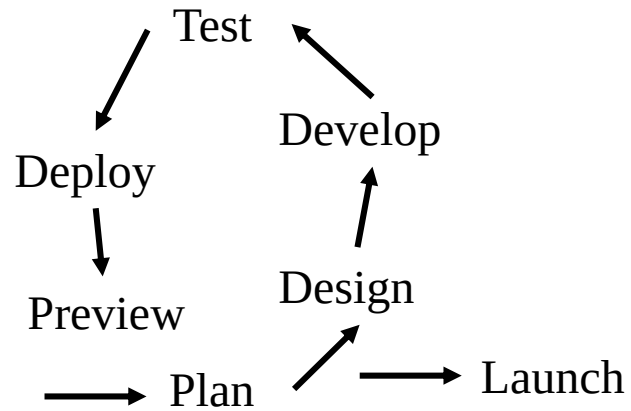


- It does not matter how many services are there. All together are shipped as a one whole application. That's how things worked previously, and this application was called monolithic application.

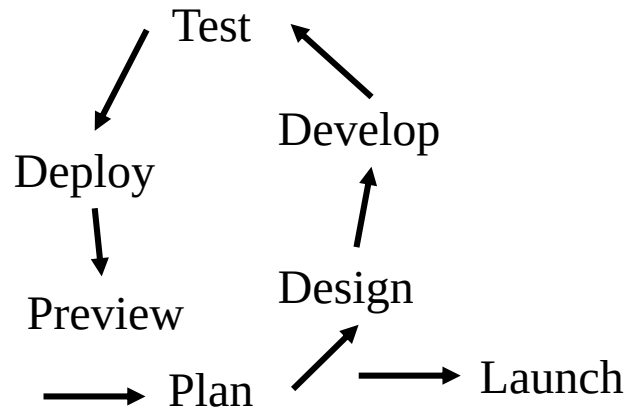
Agile

- Till 2002 it was fine.
- However, whenever one of the services is down, whole system is down.
- That's when Agile came to the picture and played an important part here.
- Agile is a philosophy to rapidly deploy an application in much more organized way.
- It is a chain of rapid development and deployment.

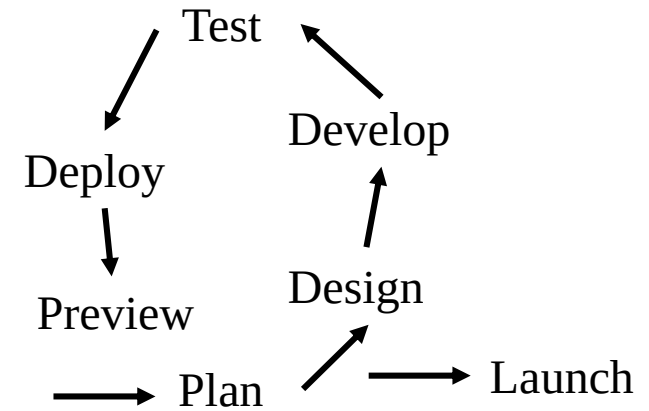
Agile



First Iteration



Second Iteration



Third Iteration

- Here, we work on one microservices at a time.
- So, in the first iteration, we will be working on a specific task, then another one in another iteration.

Key Terms of Agile

- **People over processes and tools:** A development and client centric solution.
- **Working software over comprehensive documentation:** Functional application development is more important than the documentation.
- **Customer collaboration over rigid contracts:** Agile is feedback dependent.

Principles of Agile

- Satisfy the customer.
- Welcome changing requirement.
- Deliver working software frequently.
- Frequent interaction with stakeholders.
- Motivated individuals.
- Face to face communication.
- Measure by working software.
- Maintain constant space.

Principles of Agile

- Sustain technical excellence and good design.
- Keep it simple.
- Empower self-organizing teams.
- Reflect and adjust continuously.

Advantage of Agile

- Persistent software delivery.
- Increased stakeholder satisfaction.
- Inspect and adapt.
- Welcome to changes at any stage.
- Design is important.
- Daily interaction

How to Implement Agile

- SCRUM
- Extreme Programming (XP)
- LEAN
- KANBAN
- CRYSTAL

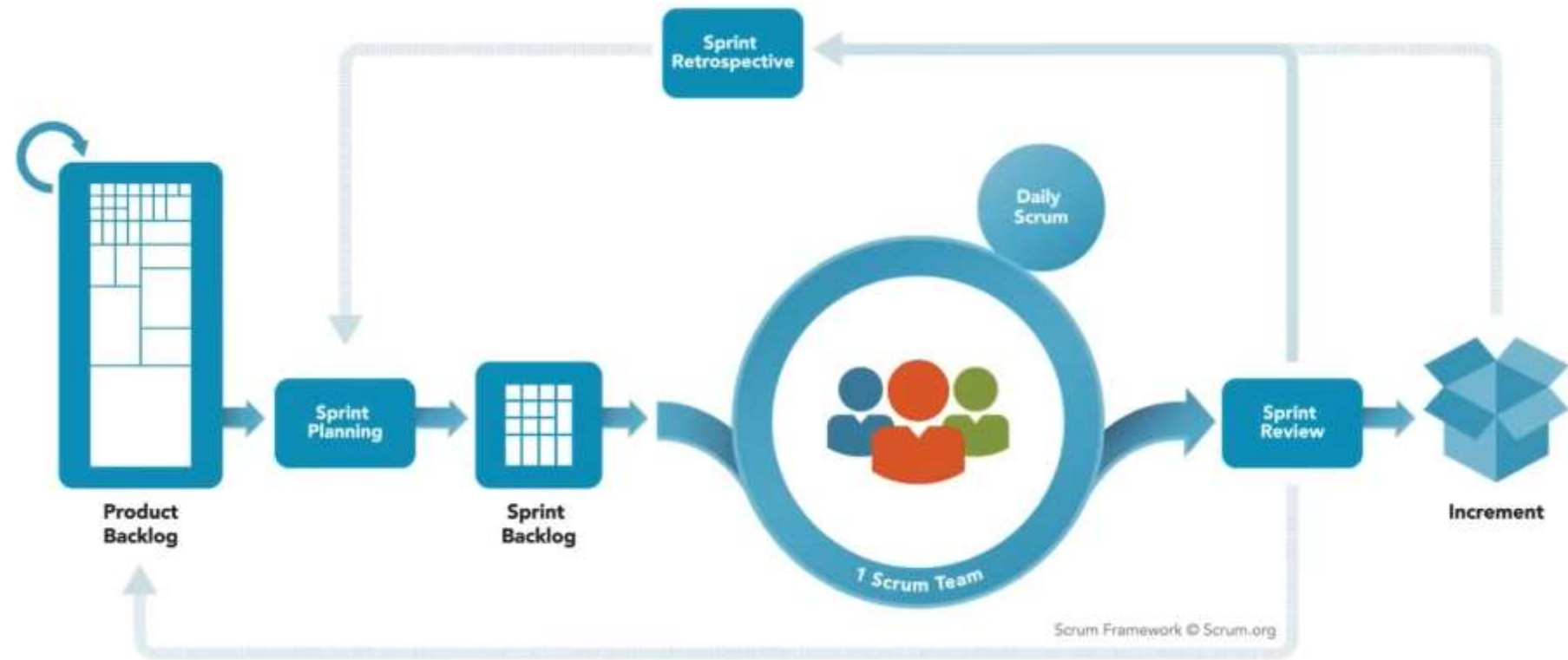
SCRUM

- An iterative approach of planning, building, testing, and reviewing.
- Here, the product owner is responsible for deployment.
- He/she doesn't need to be a technical person.
- We will have a Scrum master who is a team leader.
- Scrum master handles day to day tasks.
- Roles in Scrum:
 - Product owner
 - Scrum master
 - Team (Developer, Tester)

Development in SCRUM

- **Product Backlogs:** In the first iteration a specific task is done. Another specific task is done in the second iteration. Product owner and scrum master usually develop the product backlog.
- **User Stories:** Each task is called user stories and prioritize the product backlog tasks.
- **Sprint Backlog:** Here, development team got involved. We have **sprint planning** where product owner mentioned about the end goals or how the product will look like at the end of the day. We will have **Daily Scrum** -15 minutes stand-by meeting by developer, tester and others.

SCRUM



Development in Scrum

Sprint Reviews: At the end of every user stories[sprint backlogs], usually after 2-4 weeks there will be a demo and then move to next sprint.